Revisiting Lower Bounds for Multi-r-ic Depth Four Circuits

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Abstract

Multi-r-ic arithmetic circuits are arithmetic circuits where the individual degree of every variable, in polynomials computed at every node, is restricted to be at most r. This is a natural generalization of the multilinear restriction.

Raz and Yehudayoff (CC, 2008) introduced a "full rank polynomial" and proved strong lower bounds against the multilinear circuits (of small depth) and formulas computing it, using the method of partial derivatives (cf. Nisan-Wigderson (CC, 1997), Raz (ToC 2006)). It is a natural question to ask if these techniques and polynomial constructions be extended beyond the multilinear setting, to multi-r-ic circuit models and thus prove lower bounds against multi-r-ic circuits and in particular, depth four multi-r-ic circuits.

In this paper, we prove a superpolynomial yet quasipolynomial lower bound on the size of depth four multi-r-ic circuits computing an explicit "full rank polynomial" by just using the method of partial derivatives.

Recently Kayal, Saha and Tavenas (ToC 2018) proved an exponential lower bound on the size of depth four multi-r-ic circuits computing explicit polynomials using a stronger complexity measure. Our proof however borrows no elements from theirs and is for a completely different polynomial. The proof strategy is inspired by Saptharishi's proof of an exponential lower bound for depth three multi-r-ic circuits (cf. Chapter 14, Ramprasad's survey, 2019).

The point of this paper is to retrospectively extend our understanding of the power of the method of Partial Derivatives and thus achieve superpolynomial lower bounds, and its limits to realize the need to use stronger complexity measures to prove exponential bounds.

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1 Introduction

One of the major focal points in the area of algebraic complexity theory is to show that certain polynomials are hard to compute syntactically. Here, the hardness of computation is quantified by the number of arithmetic operations that are needed to compute the target polynomial. Instead of the standard turing machine model, we consider arithmetic circuits and formulas as models of computation.

Arithmetic circuits are directed acyclic graphs such that the leaf nodes are labeled by variables or constants from the underlying field, and every non-leaf node is labeled either by a $+$ or $\times$. Every node computes a polynomial by operating on its inputs with the operand given by its label. The flow of computation flows from the leaf to the output node. We refer the readers to the standard resources [27, 26] for more information on arithmetic formulas and arithmetic circuits.

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Valiant conjectured that the Permanent does not have polynomial sized arithmetic circuits [30]. Working towards that conjecture, we aim to prove superpolynomial circuit size lower bounds. However, the best known circuit size lower bound is $\Omega(n \log n)$, for a power symmetric polynomial, due to Baur and Strassen [28, 3], and, the best known formula size lower bound is $\Omega(n^2)$, due to Kalorkoti [15]. Due to the slow progress towards proving general circuit/formula lower bounds, it is natural to study some restricted class of arithmetic circuits and formulas.

Since most of the polynomials of interest such as Determinant, Permanent, etc., are multilinear polynomials, it is natural to consider the restriction where every intermediate computation is in fact multilinear. Due to the phenomenal work in the last two decades [20, 22, 21, 24, 25, 13, 23, 2, 7, 5, 6], the complexity of multilinear formulas and circuits is better understood than that of general formulas and circuits.

Backed with this progress it is natural to try to extend these results to a circuit model where the individual degree of every variable in the polynomial computed at every node in the circuit is r. We refer to these circuits as multi-r-ic circuits. When $r = 1$, the circuit model is multilinear.

Recently, Kumar, Oliviera and Saptharishi [18] showed that there is a chasm[1] for multi-r-ic circuits too. They proved that any polynomial sized (say n^c) multi-r-ic circuit of arbitrary depth computing a polynomial on n variables can be depth reduced to syntactical multi-r-ic depth four circuits of size $\exp(O(\sqrt{n \log n}))$. This provides us a motivation to study multi-r-ic depth four circuits and prove strong lower bounds against them. Towards this, Kayal, Saha and Tavenas [17] proved an exponential size lower bound against multi-r-ic depth four circuits computing a variant of the iterated matrix multiplication polynomial. They achieved this bound using a variant of the method *Shifted Partial Derivatives* [11] and the method of *Skew Partial Derivatives* [16] called the method of *Shifted Skew Partial Derivatives*.

Motivation for this work and our results:

For non-homogeneous multilinear circuit lower bounds (of small depth), Raz and Yehudayoff [24] introduced a "full rank polynomial" and proved lower bounds against the multilinear circuits of small depth and multilinear formulas computing it through the method of partial derivatives. It is a natural question to ask if such techniques can be extended and thus used against multi-r-ic circuits and in particular, depth four multi-r-ic circuits.

We tread this line of thought and show that we can indeed prove super polynomial size lower bounds against depth four multi-r-ic circuits computing a "full rank polynomial" by just using the method of partial derivatives.

The premise of our work is set in a hypothetical universe that is blissfully unaware of the result of Kayal, Saha and Tavenas [17] and then work towards proving lower bounds for multi-r-ic circuits. For this reason, our proof borrows no elements from [17] and is only based on tools and techniques developed prior to it.

We first construct a "full rank polynomial" (based on the construction of Saptharishi [26, Chapter 14]) and then show that any depth four multi-r-ic circuit computing it must have a super polynomial size using the dimension of partial derivatives as a complexity measure. This can be formally stated as follows.

[1] Agrawal and Vinay [1], Koiran, and Tavenas [29] showed that any general circuit can be depth reduced to a depth four circuit of non-trivial size.

▶ **Theorem 1.** *Let n, r be positive integers such that $r = o(\sqrt{\log n})$. There exists an explicit polynomial on n-variables and degree $O(n)$ such that any syntactically multi-r-ic depth four circuit computing it must be of size $n^{\Omega\left(\frac{\log n}{r^2}\right)}$.*

Comparison to related work:

It is important to note that our lower bound is quasipolynomial in the number of variables whereas the bound of Kayal, Saha and Tavenas [17] is exponential. Our result deviates from that of [17] in the following ways:

1. Our proof technique is entirely different from that of [17] and is of independent interest even in the presence of their quantitatively better bound.
2. We prove a super polynomial size lower bound against depth four multi-r-ic circuits for a new multi-r-ic full rank polynomial.
3. The polynomials considered by Kayal et al. have an implicit coding theoretic structure on a large set of their monomials making them amenable to the method of Shifted Skew Partial Derivatives. This structure is not so evident to us, in the full rank polynomial that we consider and we do not know if we can prove exponential bounds for the full rank polynomial using Shifted Skew Partial Derivatives.
4. The polynomials considered by Kayal et al. are not "full rank polynomials" and just the method of partial derivatives for those polynomials does not yield much in this multi-r-ic setting when r is not too small.

Proof overview:

Analogous to the work of Fournier et al. [9] and [19], we first consider multi-r-ic depth four circuits of low bottom support[2] and prove lower bounds against them. Let $T_1, T_2, \ldots, T_s$ be the terms corresponding to the product gates feeding into the output sum gate. The output polynomial is obtained by adding the terms $T_1, T_2, \ldots, T_s$. Recall that each of these T_i's is a product of low support polynomials $Q_{i,j}$, that is, every monomial in these $Q_{i,j}$'s is supported on a small set of variables (say μ many).

Let $\rho : X \mapsto Y \sqcup Z$ be a partition of variables X into two disjoint sets Y and Z such that each variable in X is either mapped to Y or Z with equal probability. Under such a partition, we can construct a matrix $M(f|_\rho)$ such that the rows are indexed by monomials in Y-variables and the columns are indexed by monomials in Z-variables. The entries of the matrix are defined by the coefficient of the monomial obtained by the product of its indices, in $f|_\rho$ (cf. Section 2).

As in [22, 24], we show that under a random partition ρ, for all i, $M(T_i|_\rho)$ has far from full rank with a high probability.

Let us first consider the following two extremal cases:

1. Each of the $Q_{i,j}$'s is dependent on a small set of variables: Recall that rank of a matrix is sub-multiplicative (cf. Observation 2 Item 1) and thus we get that $\text{rank}(M(T_i|_\rho)) \leq \prod_j \text{rank}(M(Q_{i,j}|_\rho))$. We first observe that a factor $Q_{i,j}$ does not contribute non-trivially to the rank if all its variables are either set only to Y-variables or only to Z-variables. In such a case $\text{rank}(M(Q_{i,j}|_\rho)) = 1$. It is easy to show that this event happens with a decent probability. We call such a factor ineffective. In particular, when a factor becomes ineffective, all the variable appearances in that factor do not contribute to

[2] That is, all the product gates at the bottom are supported on small set of variables.

the rank of $M(T_i|_\rho)$. We show that under a random partition, a good fraction of the variable appearances[3] are rendered ineffective with high probability. Conditioned on this probability, we then measure the contribution of the remaining variable appearances to the matrix rank by estimating the number of possibly non-zero rows in the matrix $M(T_i|_\rho)$. We show that this is smaller than maximum possible rank. The proof of this case borrows ideas from Saptharishi's proof [26, Chapter 14] of an exponential lower bound against depth three multi-r-ic circuits.

2. Each of the $Q_{i,j}$'s is dependent on a large set of variables: Since each of the $Q_{i,j}$'s is dependent on a large set of variables (say τ many) and total number of variable appearances is bounded by nr, we can infer that there cannot be more than nr/τ many factors. Since $Q_{i,j}$'s are polynomials with bottom support bounded by at most $\leq \mu$, $\deg(Q_{i,j})$ is at most μr and thus the degree of T_i is at most $\mu n r^2/\tau$. For a carefully chosen parameter τ, this can be small. From this, it is now clear that the possibly non-zero rows of the matrix $M(T_i|_\rho)$ will be indexed by monomials of degree at most $\mu n r^2/\tau$ in Y variables.

An arbitrary term T_i under consideration could have both types of factors, that is factors with low variable support and factors with large variable support. To deal with this, we can break our term T_i into two terms $T_{i,1}$ and $T_{i,2}$ such that $T_1 = T_{i,1} \cdot T_{i,2}$. Here, $T_{i,1}$ has factors of small support and $T_{i,2}$ has factors of large support. We then obtain rank of $M(T_{i,1}|_\rho)$ and $M(T_{i,2}|_\rho)$ separately using the aforementioned arguments and then obtain an upper bound rank of $M(T_i|_\rho)$ by multiplying ranks of $M(T_{i,1}|_\rho)$ and $M(T_{i,2}|_\rho)$ (Item 1 of Observation 2).

Once we obtain the rank of $M(T_i|_\rho)$, using sub-additivity of rank (Item 2 of Observation 2) and an union bound, we then show that the polynomial computed by a "small sized" depth four multi-r-ic circuit of low bottom support, has less than full rank for a random partition, with a non-zero probability. On the other hand, we show that an explicit candidate polynomial F_n has full rank under every partition of the variables. Hence, a depth four multi-r-ic circuit of low bottom support, of small size, cannot compute F_n. This yields us a lower bound against multi-r-ic depth four circuits of low bottom support.

Using standard techniques (cf. [26, Chapter 20]), we can then lift our lower bound against depth four multi-r-ic circuits of low bottom support to general depth four multi-r-ic circuits.

2 Preliminaries

Depth four circuits:

A depth four circuit (denoted by $\Sigma\Pi\Sigma\Pi$) over a field $\mathbb{F}$ and variables $\{x_1, x_2, \ldots, x_n\}$ computes polynomials which can be expressed in the form of sums of products of polynomials. That is, $\sum_{i=1}^{s} \prod_{j=1}^{d_i} Q_{i,j}(x_1, \ldots, x_n)$ for some d_i's. A depth four circuit of bottom support t (denoted by $\Sigma\Pi\Sigma\Pi^{\{t\}}$) is a depth four circuit where all the monomials in every polynomial $Q_{i,j}$ are supported on at most t variables.

Multi-r-ic arithmetic circuits:

We refer the readers to the standard resources [27, 26] for the definitions of arithmetic formulas and arithmetic circuits.

[3] Recall that any variable can appear at most r times among all the factors.

▶ **Definition 2** (multi-**r**-ic circuits). *Let* $\mathbf{r} = (r_1, r_2, \cdots, r_n)$.

1. *A polynomial* $f(x_1, x_2, \cdots, x_n)$ *is said to be multi-***r***-ic if for all* $i \in [n]$*, the individual degree of the variable* x_i *in the polynomial computed at every node is at most* r_i*. If* $r_1 = r_2 = \cdots = r_n = r$*, we simply refer to it as a multi-*r*-ic polynomial.*
2. *An arithmetic circuit* Φ *is said to be a multi-***r***-ic circuit if the polynomial* f_u *computed a node* $u \in \Phi$ *is a multi-***r***-ic polynomial for all* $u \in \Phi$*.*
3. *An arithmetic circuit* Φ *is said to be a syntactically multi-***r***-ic circuit if for all product gates* $u \in \Phi$ *and* $u = u_1 \times u_2 \times \cdots \times u_t$*, the total degree with respect to every variable is bounded by* r*, i.e.,* $\sum_{j\in[t]} \deg_{x_i}(f_{u_j}) \leq r$ *for all* $i \in [n]$*.*

Complexity measure:

In this article, we use the measure of the *dimension of partial derivative space* as the complexity measure to prove lower bounds (cf. [20, 22]). We follow the coefficient matrix perspective of Raz.

Let $\rho : X \mapsto Y \sqcup Z$ be a partitioning function of X variables. Any monomial m in the polynomial f over the variables X, upon the application of the partitioning function can be expressed as a product of a monomial in Y variables with a monomial in Z variables.

$$f(X) = \sum_{i\in|\mathcal{M}|} c_i \cdot m_i(X) \quad \mapsto \quad f|_\rho(Y, Z) = \sum_{i\in|\mathcal{M}|} c_i \cdot \hat{m}_i(Y) \cdot \tilde{m}_i(Z)$$

Let $M|_\rho(f|_\rho)$ be a matrix whose rows are indexed by all the monomials in Y variables and the columns are indexed by all the monomials in Z variables. The entry indexed by the monomials $(\hat{m}(Y), \tilde{m}(Z))$ in $M|_\rho(f|_\rho)$ is the coefficient of the monomial $\hat{m}(Y) \cdot \tilde{m}(Z)$ in $f|_\rho$. The complexity of f under ρ is the rank of the matrix $M_\rho(f|_\rho)$. In short, we shall denote it as $\Gamma_\rho(f|_\rho)$.

▶ **Observation 1.** *For any multi-*r*-ic polynomial* $f(X)$ *and a partitioning function* $\rho : X \mapsto Y \sqcup Z$*,* $\Gamma_\rho(f|_\rho)$ *is at most* $(r + 1)^{\min\{|Y|,|Z|\}}$*.*

▶ **Observation 2.** *Let* f_1 *and* f_2 *be two polynomials defined over the sets of variables* X*. Let* $\rho : X \mapsto Y \sqcup Z$ *be a partitioning function.*

1. ***Sub-multiplicativity:*** *If* $f(X) = f_1(X) \cdot f_2(X)$ *then* $\Gamma_\rho(f|_\rho) \leq \Gamma_\rho(f|_\rho) \cdot \Gamma_\rho(f|_\rho)$*.*
2. ***Sub-additivity:*** *If* $f(X) = f_1(X) + f_2(X)$ *then* $\Gamma_\rho(f|_\rho) \leq \Gamma_\rho(f|_\rho) + \Gamma_\rho(f|_\rho)$*.*

▶ **Observation 3.** *Let* f_1 *and* f_2 *be two polynomials defined over the disjoint sets of variables* X_1 *and* X_2*. Let* $\rho_1 : X_1 \mapsto Y_1 \sqcup Z_1$ *and* $\rho_2 : X_2 \mapsto Y_2 \sqcup Z_2$ *be two partitioning functions such that* Y_1, Y_2, Z_1, Z_2 *are all distinct. Let* $f(X_1 \cup X_2) = f_1(X_1) \cdot f_2(X_2)$ *and* $\rho = \rho_1 \oplus \rho_2$*. Then* $M_\rho(f|_\rho) = M_{\rho_1}(f_1|_{\rho_1}) \otimes M_{\rho_2}(f_2|_{\rho_2})$ *and thus,* $\Gamma_\rho(f|_\rho) = \Gamma_{\rho_1}(f|_{\rho_1}) \cdot \Gamma_{\rho_2}(f|_{\rho_2})$*.*

Useful probabilistic estimates:

The following version of the Chernoff bound [4, 12] is from the book of Dubashi and Panconesi [8, Theorem 1.1].

▶ **Theorem 3** (Chernoff bound). *Let* $W_1, \ldots, W_n$ *be independent* $\{0, 1\}$*-valued random variables and let* $W = \sum_{i\in[n]} W_i$*. Then we have the following.*

1. *For any* $\varepsilon > 0$*,*

$$\Pr[W > (1 + \varepsilon)\,\mathbb{E}[W]] \leq \exp(-\frac{\varepsilon^2}{3}\,\mathbb{E}[W]) \quad \textit{and} \quad \Pr[W < (1 - \varepsilon)\,\mathbb{E}[W]] \leq \exp(-\frac{\varepsilon^2}{2}\,\mathbb{E}[W]).$$

2. *For any* $t > 2e\,\mathbb{E}[W]$,

$$\Pr[W > t] \leq 2^{-t}.$$

We also need the following "read-r" version of concentration bounds.

▶ **Theorem 4** ([10, 14]). *Let $X_1, \ldots, X_n$ be independent random variables. Let $E_1, \ldots, E_k$ be boolean random variables that are functions of $\{X_i\}$ such that each X_i influences at most r of the E_j's. If $\Pr[E_i = 1] \geq p$, then for any $\varepsilon > 0$, we have*

$$\Pr[E_1 + \cdots + E_k \quad \leq \quad (p-\varepsilon)k] \quad \leq \quad e^{-2\varepsilon^2 k/r}.$$

3 Low-support multi-r-ic depth 4 circuits

In this section we will show that any polynomial that is computed by a syntactically multi-r-ic depth four circuit of low bottom support must have strictly less than full rank for at least one partition of the variables.

▶ **Theorem 5.** *Let $n, r > 1, \mu$ be positive integers. Let T be a syntactic multi-r-ic product of polynomials $Q_1 Q_2 \cdots Q_t$ over the variables $X = \{x_1, x_2, \ldots, x_n\}$. Let ρ be a random partition of variables X into $Y \sqcup Z$ with equal probability. Let $m = \min\{|Y|, |Z|\}$. Then, $\Gamma_\rho(T|_\rho)$ is at most $(1+r)^{m - \frac{n}{2^{40}\mu r^2 + 2}}$ with a probability of at least $1 - 2\exp\left(\frac{n}{2^{80}\mu r^2+1}\right)$.*

Proof. Since, ρ is a random partition of variables X into $Y \sqcup Z$ with equal probability, $\mathbb{E}_\rho[|Y|] = |X|/2 = n/2$. Let n_Y denote $|Y|$. Without loss of generality, let us assume that Y is the smaller partition[4]. Using Chernoff bound (from Theorem 3), we get that $n_Y \geq n/4$ with a probability of at least $1 - \exp\left(-\frac{n}{16}\right)$. From here on, we shall condition the rest of the analysis on $n_Y \geq n/4$ and use this error probability to union bound all the error probabilities at the end.

Let us further express T as a product of T_1 and T_2 where

$$T_1 = \{Q_i \mid |\mathrm{Supp}(Q_i)| < \tau\} \quad \text{and} \quad T_2 = \{Q_i \mid |\mathrm{Supp}(Q_i)| \geq \tau\}$$

for a parameter τ fixed[5] to $\frac{10\mu n r^2}{n_Y}$. Let $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ be a n-length vector that represents the maximum degree of all the variables in T_1 and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$ be a n-length vector that represents the maximum degree of all the variables in T_2. Since the circuit C is multi-r-ic , we get that $\mathbf{a} + \mathbf{b} \leq \mathbf{r}$. Further, for all $i \in [n]$, $a_i + b_i \leq r$.

From sub-multiplicativity of rank, we get that $\Gamma_\rho(T|_\rho) \leq \Gamma_\rho(T_1|_\rho) \cdot \Gamma_\rho(T_2|_\rho)$. We shall now compute $\Gamma_\rho(T_1|_\rho)$ and $\Gamma_\rho(T_2|_\rho)$ independently and thus obtain a suitable upper bound on $\Gamma_\rho(T|_\rho)$.

Computing $\Gamma_\rho(T_1|_\rho)$:

From the definition, it is clear that the support of all the polynomials in T_1 is at most τ. With some reordering of indexes, we can consider T_1 to be $Q_1 Q_2 \cdots Q_{t_1}$. For all $i \in [t_1]$, let the polynomial Q_i be defined over a variable sets X_i. Let $w_1, w_2, \ldots, w_\ell$ be an enumeration of all the appearances of the X variables in T_1 with repetition.

[4] If not, interchange the labels of the sets Y and Z.
[5] Note that technically, τ is a random variable as n_Y is a random variable.

We call a factor Q_i ($i \in [t_1]$) ineffective under ρ if all the variables in the support of Q_i are either all mapped to Y variables or to Z variables. We call a variable w_j ineffective under ρ if it appears in a factor and if it occurs in an ineffective factor. Let E_j be a $\{0,1\}$-random variable such that $E_j = 1$ if w_j is ineffective and 0 otherwise.

$$\Pr_\rho[E_j = 1] = \Pr_\rho[Q_j \text{ is ineffective under } \rho] = \frac{1}{2^{\tau-1}}.$$

Notice, however, that E_i, E_j are not independent.

Let us now count the total number of ineffective appearances of Y variables. Note that E_j's are at most r-wise dependent. This is due to the fact that any variable x_i appears in at most r many Q_j's and thus under ρ, it can influence at most r-many E_j's. Let S be a subset of $[\ell]$ such that $j \in S$ iff the X variable corresponding to w_j maps to a Y variable under ρ. Let $r_S = \sum_{i:\rho(x_i)\in Y} a_i$. Using Theorem 4 with the value of ε set to $2^{-\tau}$, we get that

$$\Pr_\rho\left[\sum_{j\in S} E_j \le \frac{r_S}{2^\tau}\right] \le \exp\left(-\frac{r_S}{r\cdot 2^{2\tau-1}}\right).$$

With probability of $\left(1-\exp\left(-\frac{r_S}{r\cdot 2^{2\tau-1}}\right)\right)$, at most $r_S\left(1-\frac{1}{2^\tau}\right)$ many variable appearances are not ineffective. Let $\mathbf{d}$ be a vector of length n that represents the effective variable appearances. From the aforementioned arguments, we get that with high probability $\sum_{i:\rho(x_i)\in Y} d_i \le r_S\left(1-\frac{1}{2^\tau}\right)$. The number of possible non-zero rows in $M_{Y,Z}(T_1)$ is at most $\prod_{i:\rho(x_i)\in Y}(1+d_i)$. Remember that $n_Y = |Y|$.

$$\begin{aligned}
\Gamma_\rho(T_1|_\rho) &\le \prod_{i:\rho(x_i)\in Y}(1+d_i) \\
&\le \left(\frac{n_Y + \sum_{i:\rho(x_i)\in Y} d_i}{n_Y}\right)^{n_Y} && \text{(Using AM-GM inequality)} \\
&\le \left(1+\frac{r_S\left(1-\frac{1}{2^\tau}\right)}{n_Y}\right)^{n_Y} \\
&\le (1+r)^{\frac{r_S}{r}\left(1-\frac{1}{2^\tau}\right)}. && \text{(Using } (1+ux)^n \le (1+x)^{un}\text{)}
\end{aligned}$$

Rephrasing this, we can say that $\Gamma_\rho(T_1|_\rho)$ is at most $(1+r)^{\frac{r_S}{r}(1-\nu)}$ where ν is at least $1/2^\tau$ with a probability of $\left(1-\exp\left(-\frac{r_S}{r\cdot 2^{2\tau-1}}\right)\right)$.

Computing $\Gamma_\rho(T_2|_\rho)$:

Under suitable renaming of polynomials, let T_2 be the product $R_1R_2\cdots R_{t_2}$. Let d_L be equal to $\sum_{i\in[n]} b_i$, r_L be equal to $\sum_{i:\rho(x_i)\in Y} b_i$. Note that $d_L \le nr$.

Trivially, the number of possible non-zero rows in $M_{Y,Z}(T_2|_\rho)$ is at most $\prod_{i:\rho(x_i)\in Y}(1+b_i)$.

$$\Gamma_\rho(T_2|_\rho) \le \prod_{i:\rho(x_i)\in Y}(1+b_i) \le \left(\frac{n_Y + \sum_{i:\rho(x_i)\in Y} b_i}{n_Y}\right)^{n_Y} \le (1+r)^{\frac{r_L}{r}}.$$

We shall now show that this bound can be improved if there is a significant chunk of Y variables that appear in T_2. From the definition, we know that for all $i \in [t_2]$, the polynomial

R_i has variable support of at least τ and every monomial depends only on μ variables[6]. This directly leads us to the following observation.

▶ **Observation 4.** *For all $i \in [t_2]$, degree of the polynomial R_i is at most μr. Thus, the degree of T_2 is at most $\mu r t_2$.*

Note that d_L counts the aggregate of the individual degrees of all the variables in T_2 and the support size of each R_i is at least τ. From this, we can infer that there cannot be *too many* factors.

▶ **Observation 5.** *The total number of factors in T_2 is at most $d_L/\tau \leq nr/\tau$. Thus, degree of T_2 is at most $\frac{\mu n r^2}{\tau}$.*

Thus, the number of potential non-zero rows in $M_{Y,Z}(T_2|_\rho)$ will be indexed by all monomials in Y of degree at most $\mu r t_2 \leq \mu n r^2/\tau$. since $\tau = \frac{10\mu n r^2}{n_Y}$, we have that $n_Y \geq \frac{\mu n r^2}{\tau} = \frac{n_Y}{10}$.

$$
\begin{aligned}
\Gamma_\rho(T_2|_\rho) &\leq \binom{n_Y + \frac{\mu n r^2}{\tau}}{n_Y} \\
&= \binom{n_Y + \frac{\mu n r^2}{\tau}}{\frac{\mu n r^2}{\tau}} \\
&\leq \left(\frac{n_Y + \frac{\mu n r^2}{\tau}}{\frac{\mu n r^2}{\tau}}\right)^{\frac{\mu n r^2}{\tau}} \cdot e^{\frac{\mu n r^2}{\tau}} && \text{(Using } \binom{n}{k} \leq (en/k)^k \text{ for } k \leq n/2\text{)} \\
&= \left(1 + \frac{n_Y \tau}{\mu n r^2}\right)^{\frac{\mu n r^2}{\tau}} \cdot e^{\frac{\mu n r^2}{\tau}} \\
&\leq (1+r)^{\frac{n_Y}{r} + \frac{\mu n r^2}{\tau \ln(1+r)}}. && \text{(Using } (1+ux)^n \leq (1+x)^{un}\text{)}
\end{aligned}
$$

We will now bound $\Gamma_\rho(T_2|_\rho)$ by $(1+r)^{\frac{r_L}{r}(1-\eta)}$ where $\eta = \max\left\{0, 1 - \frac{r}{r_L}\left(\frac{n_Y}{r} + \frac{\mu n r^2}{\tau \ln(1+r)}\right)\right\}$.

Computing $\Gamma_\rho(T|_\rho)$:

From the sub-multiplicativity of the measure, we know that $\Gamma_\rho(T|_\rho)$ is at most $\Gamma_\rho(T_1|_\rho) \cdot \Gamma_\rho(T_2|_\rho)$.

$$
\begin{aligned}
\Gamma_\rho(T|_\rho) &\leq (1+r)^{\frac{r_S}{r}(1-\nu)} \cdot (1+r)^{\frac{r_L}{r}(1-\eta)} \\
&= (1+r)^{\left(\frac{r_L + r_S}{r} - \frac{r_L \eta}{r} - \frac{r_S \nu}{r}\right)} \\
&\leq (1+r)^{\left(n_Y + \frac{r_L + r_S}{r} - \left(n_Y + \frac{r_L \eta}{r} + \frac{r_S \nu}{r}\right)\right)} \\
&= (1+r)^{n_Y - \Delta}
\end{aligned}
$$

where $\Delta = \left(n_Y + \frac{r_L \eta}{r} + \frac{r_S \nu}{r}\right) - \frac{r_L + r_S}{r}$. Note that the maximum value $\Gamma_\rho(T|_\rho)$ can attain is $(1+r)^{n_Y}$ and Δ is a measure of rank deficiency under ρ.

Recall that $r_L + r_S \leq n_Y r$ and hence $\Delta \geq 0$. Let $r_L \leq \delta n_Y r$ and $r_S = (1-\delta) n_Y r$ for some $\delta = \delta(\tau) \in [0,1]$. Note that the value of δ gets fixed given a summand T and a threshold parameter τ.

[6] Recall that each of the polynomials $R_1, R_2, \ldots, R_{t_2}$ are such that every monomial has a support of at most μ.

Case when $\delta \leq 0.75$: Since δ is small, the contribution from T_2 towards Δ might be small or nothing at all. In this case, we shall argue that Δ can be suitably lower bounded by just using the contributions from T_1 with high probability.

$$\begin{aligned}
\Delta &= \left(n_Y + \frac{r_L\eta}{r} + \frac{r_S\nu}{r}\right) - \frac{r_L + r_S}{r} \\
&\geq n_Y - \frac{r_L}{r} - \frac{r_S}{r}(1-\nu) && \text{(Since } \eta \geq 0\text{)} \\
&\geq n_Y - \delta n_Y - (1-\delta)n_Y(1-\nu) && \text{(Since } r_L \leq \delta n_Y r \text{ and } r_S = (1-\delta)n_Y r\text{)} \\
&= n_Y(1-\delta)\nu.
\end{aligned}$$

With a probability of at least $\left(1 - \exp\left(-\frac{r_S}{r \cdot 2^{2\tau-1}}\right)\right) \geq \left(1 - \exp\left(-\frac{n_Y}{2^{2\tau+1}}\right)\right)$, ν is at least $\frac{1}{2^\tau}$. Conditioned on this probability, we get that $\Delta \geq \frac{n_Y}{2^{\tau+2}}$.

Case when $\delta > 0.75$: Since δ is not too small, the contribution from T_2 towards Δ will start getting significantly larger. In this case, we shall argue that Δ can be suitably lower bounded by just using the contributions from T_2. By ignoring the contributions from T_1 to Δ, we can get rid of probabilistic arguments in this case.

Firstly, we will argue that if $\frac{r_L+r_S}{r}$ is much smaller than n_Y, then Δ is non-trivially lower bounded. Let us suppose that $\frac{r_L+r_S}{r} \leq (1-\varepsilon)\, n_Y$ for a constant ε fixed to a value of 0.1. In such a case,

$$\Delta = \left(n_Y + \frac{r_L\eta}{r} + \frac{r_S\nu}{r}\right) - \frac{r_L + r_S}{r} \geq \left(n_Y - \frac{r_L + r_S}{r}\right) \geq \varepsilon n_Y = \frac{n_Y}{10}.$$

Hereafter, we can assume that $\frac{r_L+r_S}{r} > (1-\varepsilon)\, n_Y$ and thus, $r_L \geq (\delta - \varepsilon)n_Y r$.

$$\begin{aligned}
\Delta &= \left(n_Y + \frac{r_L\eta}{r} + \frac{r_S\nu}{r}\right) - \frac{r_L + r_S}{r} \\
&\geq \left(n_Y - \frac{r_S + r_L}{r}\right) + \frac{r_L}{r} - \frac{n_Y}{r} - \frac{\mu n r^2}{\tau \ln(1+r)} \\
&\geq (\delta - \varepsilon)n_Y - \frac{n_Y}{r} - \frac{\mu n r^2}{\tau \ln(1+r)} && \text{(Since } n_Y \geq \frac{r_S + r_L}{r}\text{)} \\
&\geq \left(\delta - \varepsilon - \frac{1}{r}\right) n_Y - \frac{n_Y}{10 \ln(1+r)} && \text{(Since } \tau = 10\mu n r^2/n_Y\text{)} \\
&\geq \left(\delta - \varepsilon - \frac{1}{r} - \frac{1}{10\ln(1+r)}\right) n_Y \\
&\geq (\delta - 0.7)\, n_Y && \text{(Since } \forall\, r > 1,\ \frac{1}{r} + \frac{1}{10\ln(1+r)} < 0.6\text{)} \\
&\geq 0.05 n_Y.
\end{aligned}$$

Putting it all together:

From the above mentioned case analysis, for large values of τ we can infer that $\Delta \geq \min\left\{0.1n_Y, 0.05n_Y, \frac{n_Y}{2^{\tau+2}}\right\} \geq \frac{n_Y}{2^{\tau+2}}$ with a probability of at least $1 - \exp\left(-\frac{n_Y}{2^{2\tau+1}}\right)$. Recall that $n_Y \geq \frac{n}{4}$ with a probability of $1 - \exp\left(-\frac{n}{16}\right)$. Conditioned on this, τ is at most $40\mu r^2$. Thus, $\Delta \geq \frac{n}{2^{40\mu r^2+2}}$ with a probability of at least $1 - \left(\exp\left(-\frac{n}{16}\right) + \exp\left(-\frac{n}{2^{80\mu r^2+1}}\right)\right) \geq \left(1 - 2\exp\left(-\frac{n}{2^{80\mu r^2+1}}\right)\right)$.

This gives us the theorem statement that $\Gamma_\rho(T|_\rho)$ is at most $(1+r)^{\min\{|Y|,|Z|\}-\frac{m}{2^{40\mu r^2+2}}}$ with a probability of at least $1-2\exp\left(-\frac{n}{2^{80\mu r^2+1}}\right)$. ◀

▶ **Theorem 6.** *Let n, r, μ be positive integers. Let C be a syntactically multi-r-ic depth four $\Sigma\Pi\Sigma\Pi^{\{\mu\}}$ circuit of size $s < \exp\left(\frac{n}{2^{80\mu r^2+2}}\right)$ that computes a multi-r-ic polynomial f over the variables $X = \{x_1, x_2, \cdots, x_n\}$. Then there exists a random partition ρ of variables X into $Y \sqcup Z$ with equal probability such that $\Gamma_\rho(f|_\rho)$ is strictly less than $(1+r)^{\min\{|Y|,|Z|\}}$.*

Proof. Recall that f is a sum of at most s many products of polynomials $T^{(1)} + \cdots + T^{(s)}$. Using sub-additivity of measure, we get that $\Gamma_\rho(f|_\rho) \leq s \cdot \max_{i\in[s]} \Gamma_\rho(T^{(i)}|_\rho)$. Let $m = \min\{|Y|, |Z|\}$. From Theorem 5, we get that for a term $T^{(i)}$, $\Gamma_\rho(T^{(i)}|_\rho)$ is at most $(1+r)^{m-\frac{n}{2^{40\mu r^2+2}}}$ with a probability of at least $1-2\exp\left(-\frac{n}{2^{80\mu r^2+1}}\right)$. Taking a union bound over all the $T^{(i)}$'s, we get that for all $i \in [s]$, $\Gamma_\rho(T^{(i)}|_\rho)$ is is at most $(1+r)^{m-\frac{n}{2^{40\mu r^2+2}}}$ with a probability of at least $p = 1-2s\exp\left(-\frac{n}{2^{80\mu r^2+1}}\right)$. Since $s < \exp\left(\frac{n}{2^{80\mu r^2+2}}\right)$, p is strictly greater than zero.

Thus with a non-zero probability, we get that $\Gamma_\rho(f|_\rho)$ is strictly less than $(1+r)^m$.

$$\begin{aligned}
\Gamma_\rho(f|_\rho) &\leq s \cdot \max_{i\in[s]} \left\{\Gamma_\rho(T^{(i)}|_\rho)\right\} \\
&\leq s \cdot (1+r)^{m-\frac{n}{2^{40\mu r^2+2}}} \\
&< (1+r)^{m-\frac{n}{2^{40\mu r^2+2}}+\frac{n}{2^{80\mu r^2+2}\log(1+r)}} \\
&< (1+r)^m .
\end{aligned}$$

◀

4 Multi-r-ic depth 4 circuits

Before we describe the proof of the lower bound, we shall introduce our polynomial family. To begin with, we consider a polynomial which is a slight generalization of a *full rank* polynomial of Raz and Yehudayoff [24][7]. Using this polynomial, we construct the hard polynomial F by composing the full rank polynomial with linear forms akin to [26].

The full rank polynomial F_n is defined over the field $\mathbb{F}(\{w_{a,b,c}, w_{a,b} : a, b, c \leq 2n\})[x_1, x_2, \ldots, x_n]$. In short, we denote it as $\mathbb{F}(W)[X]$. Notice, that $w_{a,b,c}$ and $w_{a,b}$ are some auxiliary variables here. The full rank polynomial $F_n = f_{1,2n}$ is defined recursively as follows.

$$f_{i,j} = \begin{cases} \sum_{k=0}^{r} x_i^k x_{i+1}^k & \text{if } j = i+1, \\ \left(\sum_{k=0}^{r} x_i^k x_j^k\right) \cdot f_{i+1,j-1} \cdot w_{i,j} & \\ \quad + \sum_{\ell=i+1}^{j-1} f_{i,\ell} \cdot f_{\ell+1,j} \cdot w_{i,\ell,j} & \text{if } j-i \text{ is odd}, \\ 0 & \text{otherwise.} \end{cases}$$

[7] This was first defined and was then used by Saptharishi to prove a multi-r-ic depth three lower bound [26]

We can now define our hard polynomial P_n based on the definition of F_n. Let $p = n^{-c}$ for some constant $c \in (0,1)$ which we shall later fix. Let $\tilde{X} = \{\tilde{x}_1, \ldots, \tilde{x}_{2n}\}$ and $\hat{X} = \{\hat{x}_{1,1}, \hat{x}_{1,2}, \ldots, \hat{x}_{1,t}, \ldots, \hat{x}_{2n,1}, \hat{x}_{2n,2}, \ldots, \hat{x}_{2n,t}\}$ be distinct variable sets, where $t = \frac{2n\log(2n)}{p}$. Let $\mathrm{Lin}_p : \tilde{X} \mapsto \hat{X}$ be a linear map such that $\tilde{x}_i \mapsto \sum_{j=1}^{t} \hat{x}_{i,j}$ for all $i \in [2n]$. The polynomial $P_{n,p}(\hat{X})$ be defined to be $F_n \circ \mathrm{Lin}_p(\hat{X})$. That is,

$$P_{n,p} = F_n(\sum_{j=1}^{t} \hat{x}_{1,j}, \sum_{j=1}^{t} \hat{x}_{2,j}, \cdots, \sum_{j=1}^{t} \hat{x}_{2n,j}) \quad \text{where } t = \frac{2n\log(2n)}{p}.$$

We will now recall the following useful trick due to Kumar and Saptharishi [26].

▶ **Lemma 7** (Lemma 20.5 [26]). *Let σ be a random restriction on the variable set $\hat{X}$ that sets a variable to zero with a probability of $(1-p)$ where $p = n^{-c}$ for some constant $c \in (0,1)$. Then F_n is a projection of $(P_{n,p})|_\sigma$ with a probability of at least $(1 - 2^{-n})$.*

We will now show that the matrix corresponding to the polynomial F_n under every partition that we consider is of full rank and hence it is a *hard* polynomial for the circuit model under consideration.

▶ **Lemma 8.** *The polynomial $F_n = f_{1,2n}$ defined above has the property that for every partition $X = \{x_1, \ldots, x_{2n}\} = Y \sqcup Z$, we have*

$$\Gamma_\rho(F_n|_\rho) \quad \geq \quad (r+1)^{\min(|Y|,|Z|)}.$$

Further, this polynomial can be computed by a linear sized arithmetic circuit and hence is in VP.

Proof. The proof is similar to Lemma 13.17 [26]. We use an induction over n, where $2n$ is the number of variables. For $n = 1$ we have the polynomial $F_1 = \sum_{k=0}^{r} x_1^k x_2^k$. It is clear that any partition of x_1 and x_2 will have rank of $(r+1)^{\min(|Y|,|Z|)}$. That is, $\Gamma_\rho(F_1) = 1$ if both the variables are on the same side, and is equal to $r+1$ if there is an equi-partition.

Let $n > 1$ and let $\rho : X \mapsto Y \sqcup Z$ be any partition. Under the partition ρ, the variables x_1, x_{2n} could either be to the same side or get mapped to different sets, amongst Y and Z. We handle these cases separately.

Let $\hat{Y} = Y \setminus \{\rho(x_1), \rho(x_{2n})\}$ and $\hat{Z} = Z \setminus \{\rho(x_1), \rho(x_{2n})\}$.

Case 1: x_1, x_{2n} are in different parts.

For ease of argument we set $w_{1,i,2n} = 0$ for all i and let $w_{1,2n} = 1$. Then we see that our polynomial is given by $\sum_{r=0}^{k} (x_1^r x_{2n}^r)\, f_{2,2n-1} w_{i,j}$. By invoking the induction hypothesis, we get that $f_{2,2n-1}$ has a rank of at least $(r+1)^{\min(|\hat{Y}|,|\hat{Z}|)}$. Since $\sum_{r=0}^{k} (x_1^r x_{2n}^r)$ and $f_{2,2n-1}$ are defined over disjoint sets of variables, Observation 3 leads us to the following.

$$\begin{aligned}\Gamma_\rho(\sum_{r=0}^{k} (x_1^r x_{2n}^r)\, f_{2,2n-1}) &= \Gamma_\rho(\sum_{r=0}^{k} (x_1^r x_{2n}^r)) \cdot \Gamma_\rho(f_{2,2n-1}) \\ &\geq (r+1) \cdot (r+1)^{\min(|\hat{Y}|,|\hat{Z}|)} \\ &= (r+1)^{\min(|Y|,|Z|)}.\end{aligned}$$

Case 2: x_1, x_{2n} are in the same part. We know that there exists some k such that $[1,k]$ and $[k+1, 2n]$ have equal size. We set $w_{1,2n}$ to zero. We can use our induction hypothesis on $f_{1,k}$ and $f_{k+1,2n}$. Let us define $Y_{1,k} = \{\rho(x_1), \rho(x_2), \ldots, \rho(x_k)\} \cap Y$ and $Z_{1,k} = \{\rho(x_1), \rho(x_2), \ldots, \rho(x_k)\} \cap Z$. Similarly, we can define $Y_{k+1,2n} =$

$\{\rho(x_{k+1}), \rho(x_{k+2}), \ldots, \rho(x_{2n})\} \cap Y$ and $Z_{k+1,2n} = \{\rho(x_{k+1}), \rho(x_{k+2}), \ldots, \rho(x_{2n})\} \cap Z$. By invoking the induction hypothesis, we get that $\Gamma_\rho(f_{1,k}) \geq (r+1)^{\min\{|Y_{1,k}|,|Z_{1,k}|\}}$ and $\Gamma_\rho(f_{k+1,2n}) \geq (r+1)^{\min\{|Y_{k+1,2n}|,|Z_{k+1,2n}|\}}$. From the definition of the polynomial, $f_{1,k}$ and $f_{k+1,2n}$ are defined over disjoint sets of variables and thus using Observation 3 we get that

$$\Gamma_\rho(f_{1,2n}) \geq (r+1)^{\min\{|Y_{1,k}|,|Z_{1,k}|\}} \cdot (r+1)^{\min\{|Y_{k+1,2n}|,|Z_{k+1,2n}|\}} \geq (r+1)^{\min(|Y|,|Z|)}.$$

The last inequality follows from the fact that $\min\{a_1, b_1\} + \min\{a_2, b_2\} \geq \min\{a_1 + a_2, b_1 + b_2\}$. It is easy to construct a polynomial sized circuit for F_n bottom-up by following the recursive definition and infer that it is in fact in VP. ◀

For the moment, let us assume that the polynomial $F_{n/2}$ is being computed by a multi-r-ic depth four $\Sigma\Pi\Sigma\Pi^{\{\mu\}}$ circuit of size $s < \exp\left(\frac{n}{2^{80\mu r^2+2}}\right)$. Theorem 8 tells us that for all partitions $\rho : X \mapsto Y \sqcup Z$ that we consider, $\Gamma_\rho(F_{n/2}|_\rho)$ is at least $(1+r)^{\min\{|Y|,|Z|\}}$. On the other hand, we infer from Theorem 6 there is at least one partition ρ such that $\Gamma_\rho(F_{n/2}|_\rho)$ is strictly less than $(1+r)^{\min\{|Y|,|Z|\}}$ and thus arriving at a contradiction to our assumption. This can formally be summarized as follows.

▶ **Theorem 9.** *Let n, r, μ be positive integers. Any syntactically multi-r-ic depth four $\Sigma\Pi\Sigma\Pi^{\{\mu\}}$ circuit that computes $F_n(x_1, x_2, \ldots, x_{2n})$ must be of size $\exp\left(\Omega\left(\frac{n}{2^{80\mu r^2}}\right)\right)$.*

To prove the main theorem, we also need the following lemma.

▶ **Lemma 10** (Analogous to Lemma 20.4 [26]). *Let P be a multi-r-ic polynomial that is computed by a syntactically multi-r-ic depth 4 circuit C of size $s \leq n^{\delta\mu}$ for some $\delta > 0$. Let σ be a random restriction that sets each variable to zero independently with probability $(1 - n^{-2\delta})$. Then with probability at least $(1 - 1/s)$ the polynomial $\sigma(P)$ is computed by a multi-r-ic depth four circuit C' of bottom support at most μ and size s.*

Equipped with Theorem 10 and Theorem 9, we are now ready to prove our main theorem.

▶ **Theorem 11.** *Let n, r be positive integers and δ, c be some constants in $(0, 1)$. Let the parameter p be equal to $n^{-\delta}$. If C be any syntactically multi-r-ic depth four circuit computing the polynomial $P_{n,p}(\tilde{X})$ as described above then it must be of size $n^{\Omega\left(\frac{\log n}{r^2}\right)}$.*

Proof. Let μ be a parameter that is set to $\frac{\gamma \log n}{r^2}$ for a small constant γ. Let $\sigma : \hat{X} \mapsto \{0, *\}$ be a random restriction such that a variable is set to zero with a probability of $(1 - n^{-2\delta})$, and is left untouched otherwise. Let C be a syntactically multi-r-ic depth four circuit of size $s \leq n^{\delta\mu}$ that computes $P_{n,p}$. Theorem 10 tells us that $C' = \sigma(C)$ is a multi-r-ic depth four circuit of size s and bottom support at most μ. Hence, $\sigma(P_{n,p})$ has a multi-r-ic $\Sigma\Pi\Sigma\Pi^{\{\mu\}}$ size at most s. Since F_n is a p-projection of $\sigma(P_{n,p})$, F_n also has a multi-r-ic $\Sigma\Pi\Sigma\Pi^{\{\mu\}}$ of size at most s.

On the other hand, from Theorem 9 we know that any multi-r-ic $\Sigma\Pi\Sigma\Pi^{\{\mu\}}$ circuit that computes F_n must be of size $\exp\left(\Omega\left(\frac{n}{2^{80\mu r^2}}\right)\right)$. Thus, it must be that $\exp\left(\Omega\left(\frac{n}{2^{80\mu r^2}}\right)\right) \leq s \leq n^{\delta\mu}$. We can choose γ to be a small enough constant such that the aforementioned expression is satisfied. Thus, s must at least be $n^{\Omega\left(\frac{\log n}{r^2}\right)}$. ◀

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